Alexandra Ding

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EDUCATION

Massachusetts Institute of Technology **Cambridge, MA**

B.S. in Computer Science and Engineering September 2024 – expected May 2028

GPA: 5.0

Relevant Coursework: Interpreters, Efficient AI, Dynamics and Controls, Software Construction, Computer Systems, Algorithms II, C and Assembly, Processor Design, Linear Algebra, Differential Equations

SKILLS

- **Programming:** Python (OpenCV, multiprocessing, matplotlib, pytest, pandas, PyTorch), Typescript, C++, ROS, C, CSS, HTML, Java, Bash, Minispec, RISC-V assembly, Git, Linux
- **Analysis & CAD:** Statics, FEA, CFD (SimScale, ParaView), XFLR5**,** Fusion 360, Onshape, KiCAD

WORK EXPERIENCE

Relativity Space **Long Beach, CA**

Robotics Software Engineer Intern May 2026 – August 2026

- Engineered a 60Hz real-time computer vision rigid body tracking system in C++ and ROS2, built a custom multi-threaded architecture to optimize concurrency
- Integrated robust computer vision algorithms (geometric hashing, RANSAC) and an Error State Kalman Filter for next-frame pose prediction to minimize latency
- Developed Python validation tools to verify the accuracy of a multi-camera photogrammetry rig, proving sub-millimeter precision and system reliability over multi-hour continuous operations

MIT Materials Research Laboratory **Cambridge, MA**

Student Worker - AI Tools Development January 2026 – present

- Architected an automated data pipeline to identify the research needs of 500+ companies
- Utilized web scrapers and APIs to ingest data from thousands of academic papers, job postings, and patents; extracted and vectorized keywords with sentence transformers; stored data in SQLite3 database
- Operationalized the platform to rapidly match inbound corporate technical queries with relevant researchers and generate targeted outreach leads for MIT faculty

Taşan Group **Cambridge, MA**

Undergraduate researcher September 2025 – December 2025

- Fine-tuned Mistral 7B Instruct v0.3 using LoRA to automate initial feedback on scientific drafts
- Deployed LLM locally using Hugging Face tools to maintain data privacy for unpublished research

NASA Ames Unitary Plan Wind Tunnel **Mountain View, CA**

Intern June 2025 – August 2025

- Characterized the uncertainty of a computer vision angle of attack measurement system using Monte Carlo
- Developed and optimized a Python pipeline using OpenCV and multiprocessing to separately quantify error from detection and from calibration across thousands of large-format images

MIT Sea Grant **Cambridge, MA**

Undergraduate researcher October 2024 – May 2025

- Designed, fabricated, and assembled custom PCBs for an autonomous oyster-basket flipping robotic boat, integrating a Raspberry Pi, Pixhawk flight controller, and I^2C-to-PWM breakout boards

EXTRACURRICULAR ACTIVITIES

Design Build Fly **Cambridge, MA**

Aerodynamics subteam lead September 2024 – May 2026

- Designed and CADed wing for 2026 competition plane, creating a parametric, reusable design and emphasizing manufacturability